

→ we now repeat a lot of what we did ~~for~~ⁱⁿ DT for CT

→ Nearly all concepts, definitions etc. remain same!

$$DT \rightarrow CT \left\{ \begin{array}{l} \rightarrow n \rightarrow t \quad [\text{time domain}] \\ \rightarrow \mathbb{Z}^n \rightarrow e^{st} \quad [s = \sigma + j\omega] \quad [\text{freq. domain}] \\ \rightarrow \sum \rightarrow \int \quad [\text{operations}] \end{array} \right.$$

Systems → Definition & ways of checking LTI remain same.

Signals → IMPULSE DT

$$s[n] = \begin{cases} 1 & n=0 \\ 0 & \text{o/w} \end{cases}$$

$$s[n-n]x[n] = s[n-n]x[n]$$

$$\sum_{k=-\infty}^{\infty} x[k] s[n-k] = x[n]$$

$$x[n] * s[n] = x[n]$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] s[n-k]$$

↑
sum of scaled & shifted input seq.

CT

$$s(t) \text{ s.t. } s(t) = 0 \quad t \neq 0 \\ \text{and } \int_{-\infty}^{\infty} s(t) dt = 1$$

$$s(t-\tau)\phi(t) = s(t-\tau)\phi(\tau)$$

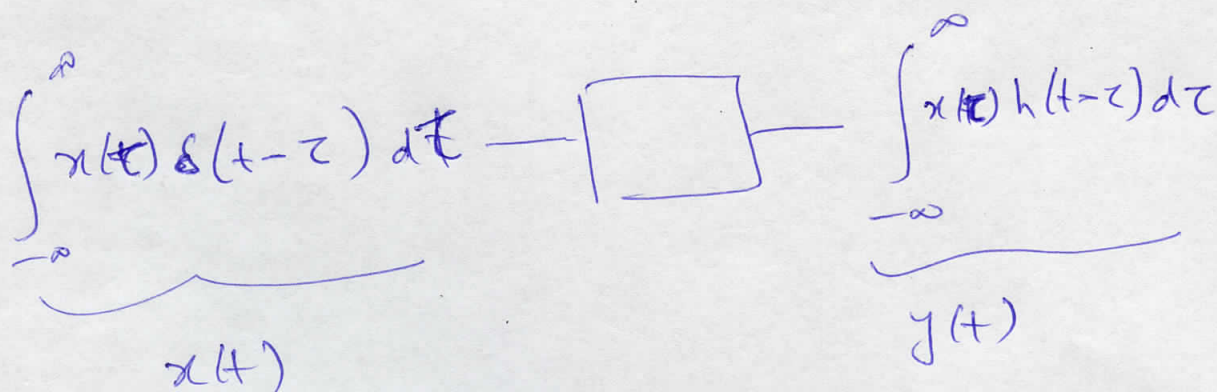
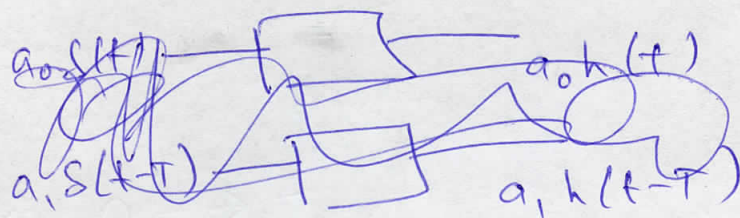
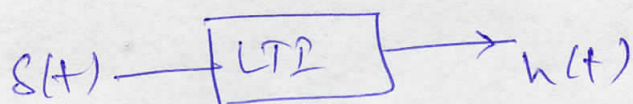
$$\int_{-\infty}^{\infty} \phi(t) s(t-\tau) dt = \phi(\tau)$$

$$x(t) * s(t) = x(t)$$

$$x(t) = \int_{-\infty}^{\infty} x(\tau) s(t-\tau) d\tau$$

→ LTI RR

(2)



$$\Rightarrow y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \triangleq x(t) * h(t)$$

→ conv. properties similar to DT case (see Lathi 2.4.1 & 2.4.2)

→ characterization of system via IR

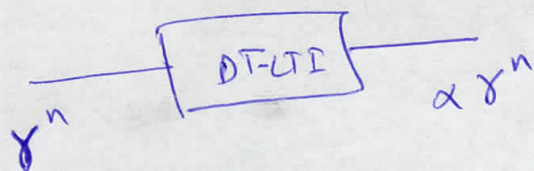
also similar to DT case.

— e.g. causal iff $h(t) = 0 \quad t < 0$

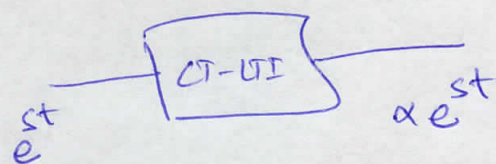
— BIBO stable iff $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

— etc.

→ Frequency Domain



eigen functions
(x complex)



(s complex)

→ $s = \sigma + j\omega \Rightarrow e^{st} = e^{(\sigma + j\omega)t} = e^{\sigma t} e^{j\omega t}$

→ Real and complex parts fixed/damped/growing sinusoids.

→ $z^{Tx} \rightarrow$ Laplace Tx.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Bilateral
Z-Tx

★ → $X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$

Bilateral
Laplace Tx.

$$e^{st} = e^{\sigma t} e^{j\omega t}$$

→ Again, what is (s)? $s = \sigma + j\omega$

→ ROC: Does $X(s)$ exist for any selection of s ?

→ for a given signal $x(t)$ we can only choose values of s for which

★ converges!

related problem
(practice later)

→ Properties & Tx-Tables

→ similar to Z-Tx.

→ Unilateral Laplace Tx.

$$X(s) = \int_0^{\infty} x(t) e^{-st} dt$$

(avoids some issues of inverse Tx)
faced by bilateral

→ Laplace Tx examples.

$$\text{— } \mathcal{L}[s(t)] = \int_0^{\infty} s(t) e^{-st} dt$$

$$= e^{-st} \Big|_{t=0}^{\infty} \int_0^{\infty} s(t) dt = 1$$

(ROC)
converges
for all s

$$s(t) \iff 1$$

$$\text{— } \mathcal{L}[u(t)] = \int_0^{\infty} u(t) e^{-st} dt = \int_0^{\infty} e^{-st} dt$$

$$= -\frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$$

(ROC)
 $\text{Re } s > 0$

for $e^{-(\text{Re } s)t}$
to vanish

$\text{Re } s > 0$

$$\text{— } \mathcal{L}[e^{-at} u(t)] = \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt$$

$$= -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty} = \frac{1}{s+a}$$

(ROC)
 $\text{Re } s > -a$

(5)

→ some properties.

→ Linear. if $x_1(t) \Leftrightarrow X_1(s)$
 $x_2(t) \Leftrightarrow X_2(s)$

$$\text{then } a_1 x_1(t) + a_2 x_2(t) \Leftrightarrow a_1 X_1(s) + a_2 X_2(s)$$

→ Time shifting if $x(t) \Leftrightarrow X(s)$
 $x(t-t_0) \Leftrightarrow X(s)e^{-st_0}$
 then

→ Freq. shifting.

if $x(t) \Leftrightarrow X(s)$
 then $x(t)e^{s_0 t} \Leftrightarrow X(s-s_0)$

→ Differentiation (time) property

if $x(t) \Leftrightarrow X(s)$

then $\frac{dx(t)}{dt} \Leftrightarrow sX(s) - \underbrace{x(0)}_{\text{initial condition}}$

and $\frac{d^2 x(t)}{dt^2} \Leftrightarrow s^2 X(s) - s \underbrace{x(0)}_{\text{initial condition}} - \underbrace{\dot{x}(0)}_{\text{initial condition}}$

etc.

→ Integration (time) property

$$\text{if } x(t) \Leftrightarrow X(s)$$

$$\text{then } \int_0^t x(z) dz \Leftrightarrow \frac{X(s)}{s}$$

→ Scaling (time)

$$\text{if } x(t) \Leftrightarrow X(s)$$

$$\text{then } x(at) \Leftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right)$$

→ Convolution (time & freq).

$$\text{if } x_1(t) \Leftrightarrow X_1(s)$$

$$x_2(t) \Leftrightarrow X_2(s)$$

$$\text{then } x_1(t) * x_2(t) \Leftrightarrow X_1(s) X_2(s)$$

$$\text{and } x_1(t) x_2(t) \Leftrightarrow \frac{1}{2\pi j} [X_1(s) * X_2(s)]$$

→ Initial & final value of $x(t)$

$$x(0^+) = \lim_{s \rightarrow \infty} s X(s)$$

$$x(\infty) = \lim_{s \rightarrow 0} s X(s)$$